

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%
Software: Cisco Packet Tracer, Wireshark

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

1. IPv4 Network Debugging

Given

- IP: 192.168.10.65
- Mask: 255.255.255.192 (/26)
- Gateway: 192.168.10.1

a) Subnet

192.168.10.64/26

b) Network Details

- Network Address: 192.168.10.64
- Broadcast Address: 192.168.10.127
- Host Range: 192.168.10.65 – 192.168.10.126

c) Verify

- IP Address: Valid
- Gateway: Invalid (different subnet)

d) Error

Gateway is in another subnet.

e) Correct Configuration

- IP: **192.168.10.65**
- Gateway: **192.168.10.66** (or any valid IP in the same subnet)

f) Usable Hosts

62 Hosts

2. IPv6 Debugging

Given

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

a) Expanded Address

Device A

2001:0db8:0000:0000:0000:ff00:0042:8329

Device B

2001:0db8:0000:0000:0001:0000:0000:0001

b) Network Prefix

Both: **2001:db8:0:0::/64**

c) Same Subnet?

Yes

d) Error

No addressing error. Check gateway, cable, or firewall.

e) Correct Example

- Device A: **2001:db8:0:0::10/64**
- Device B: **2001:db8:0:0::20/64**
- Gateway: **2001:db8:0:0::1**

f) Available Hosts

2⁶⁴ addresses

3. Subnet Analysis

Subnet	Network	Broadcast	Host Range	Usable Hosts
/26	192.168.1.0	192.168.1.63	192.168.1.1–62	62
/27	192.168.1.64	192.168.1.95	192.168.1.65–94	30
/28	192.168.1.96	192.168.1.111	192.168.1.97–110	14

Problems

- /26 may waste IPs.
- /28 may not have enough IPs.

Solution

Use **VLSM**:

- Large department → /26
- Medium department → /27
- Small department → /28

4. IPv6 Debugging

Given

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

a) Expanded Address

Device A

2001:0db8:0000:0000:0000:ff00:0042:8329

Device B

2001:0db8:0000:0000:0001:0000:0000:0001

b) Network Prefix

Both: **2001:db8:0:0::/64**

c) Same Subnet?

Yes

d) Error

No addressing error. Check gateway, cable, or firewall.

e) Correct Example

- Device A: **2001:db8:0:0::10/64**
- Device B: **2001:db8:0:0::20/64**
- Gateway: **2001:db8:0:0::1**

f) Available Hosts

2⁶⁴ addresses

- Network Prefix: **2001:db8:0:0::/64**
- Same Subnet: **Yes**
- Correct Addresses:
 - Device A → **2001:db8:0:0::10/64**
 - Device B → **2001:db8:0:0::20/64**
- Available Hosts: **2⁶⁴**

5. Routing Debugging

Routing Table

Destination	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default	10.0.0.1

a) Matching Route

192.168.2.0/24

b) Next Hop

10.0.0.3

c) Destination Check

192.168.2.100 belongs to **192.168.2.0/24**

d) Possible Error

- Wrong next hop
- Interface down
- Missing route

e) Optimization

Keep correct routes and verify the next-hop router is reachable.

f) If No Match Exists

Use the **Default Route** → **10.0.0.1**.